UF Policy RAG Chatbot

Executive Summary

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1. Problem Definition

Students often describe policy questions in everyday language, while university documents use formal terms, detailed conditions, and exceptions. This difference can make the correct section difficult to find. The University of Florida (UF) Regulation and Policy Hub [1] provides an official source, but a general chatbot may answer without checking it. A standard retrieval-augmented generation (RAG) system can supply evidence, yet its pretrained retriever may still miss relevant sections. This project will investigate whether training the retriever on student-style questions improves policy retrieval and the support behind generated answers.

2. Proposed Solution

The project will build a UF policy chatbot with a **BGE embedding model fine-tuned by the student**. The system will retrieve relevant sections and use a GPT model to generate an answer with supporting passages and source links. A Streamlit interface will display the question, answer, and evidence. When the collection cannot support an answer, the chatbot should state that limitation.

Research question: Does fine-tuning BGE on synthetic student-style questions improve retrieval of relevant UF policy sections compared with BM25 keyword search and pretrained BGE, particularly when questions use different wording from the source?

Hypothesis: BGE fine-tuned on student-style question–policy pairs will retrieve relevant sections more accurately than pretrained BGE, especially for questions with limited vocabulary overlap with the policy. This is a hypothesis to test, not an assumed result.

3. Data Plan

The source dataset will cover **all publicly accessible policy and regulation documents listed in the UF hub**. The material already exists online, but the project collection still needs to be assembled and reviewed. It is appropriate because official policy sections provide identifiable evidence for both training and evaluation. The final size and any inaccessible documents will be reported. No private student records are required.

Training data will consist of synthetic everyday questions paired with the policy sections that answer them. NaviGator will generate candidate questions, followed by checks for answerability, duplicates, and incorrect source matches. Policies will be divided into training, development, and test groups before question generation. Development and test policies will remain searchable, but their questions will not be used for training.

An independent final test set will contain approximately **150 manually written or independently reviewed questions: 120 answerable and 30 unanswerable from the collected corpus**. Answerable questions will identify all acceptable supporting sections. The set will include direct questions, paraphrases, and questions requiring several sections. A separate development set will support model selection and refusal settings; the final test set will remain untouched until evaluation.

[1] UF Regulation and Policy Hub: `https://policy.ufl.edu/`.

4. Technology and Approach

The planned stack is **Python, Beautiful Soup, Sentence Transformers, PyTorch, BGE, ChromaDB, BM25, the UF NaviGator AI Toolkit API, and Streamlit**. A GPT model accessed through NaviGator [2] will support synthetic question generation and evidence-based answers. Streamlit will be the only application interface.

The initial model will be `BAAI/bge-small-en-v1.5`, fine-tuned with a contrastive loss such as MultipleNegativesRankingLoss.[3] Training will increase similarity for matching question–section pairs relative to unrelated pairs. Duplicate positive passages within a batch will be avoided. Checked hard negatives, meaning similar but irrelevant sections, will be an optional extension. Policy embeddings will then be regenerated in a separate ChromaDB collection.

The experiment will compare **BM25, pretrained BGE, fine-tuned BGE, and BM25 combined with fine-tuned BGE**. The hybrid will combine their rankings using reciprocal rank fusion. All systems will use the same corpus, section boundaries, test questions, number of returned passages, GPT model, and answer prompt. Retrieval will be measured using Recall@5 and mean reciprocal rank on answerable questions. Manual evaluation will assess answer correctness, appropriate refusal, and **citation support: the proportion of cited claims actually supported by their cited passages**. Missing citations for substantive claims will also be recorded. A correct URL alone will not count as supporting evidence.

5. Innovation

The contribution is a controlled study of adaptation from everyday student language to formal UF policy language. Fine-tuning BGE is established; this project will test its value for UF policies. Results will distinguish questions with high and low vocabulary overlap and examine whether retrieval gains improve citation support, demonstrating the trained component's contribution beyond a standard RAG setup.

6. Anticipated Challenges and Risk Mitigation

1. Synthetic data quality and evaluation leakage. Generated questions may be incorrect, repetitive, or too similar to policy wording. Source checks, varied question styles, duplicate removal, policy-level splits, and independent test questions will reduce these risks. Self-authored evaluation remains a limitation.

2. Overfitting and misleading negatives. Fine-tuning may fail to improve retrieval, and related sections may both be valid answers. Development-based model selection, checks on negative examples, and comparison with pretrained BGE will guide training. Multiple valid sections will be annotated where appropriate, and negative results will be reported.

3. Time, compute, and access. GPU availability, API limits, and manual review may constrain progress. Access will be checked early, a compact BGE model and a limited training pilot will be used, and hard-negative experiments will remain optional. A dated corpus and extraction checks will reduce errors from missing or revised documents.

7. Potential Impact and Feasibility

Students may benefit from clearer policy information and verifiable evidence. Existing public documents, a compact model, and one Streamlit interface keep the semester scope manageable, subject to early confirmation of compute and API access. Work will cover data preparation, baselines and fine-tuning, interface integration, and final testing. Deliverables will include the documented corpus, training pairs, separate test set, fine-tuned model, chatbot, and comparative results. Official documents remain authoritative; university-wide deployment is beyond the semester scope.

[2] UF NaviGator AI Toolkit: `https://docs.ai.it.ufl.edu/docs/navigator_toolkit/getting_started/quickstart/`.
[3] Training resources: BGE model card and Sentence Transformers loss documentation.